

Vitaly Kurnosenko

Senior Backend Engineer | Python · FastAPI · AWS & GCP · APIs · IaC · CI/CD · Observability

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Professional Summary

Senior engineer with a **strong formal CS background (PhD, mathematical sciences)** and years of **production ownership** across **distributed backends**, **data-heavy services**, and **cloud infrastructure**. Comfortable owning the full path from **API design** and **service boundaries** through **IaC**, **CI/CD**, **runtime operations**, and **incident response**.

Most recent work spans **international e-commerce / fintech** (Yofi - anti-fraud and customer intelligence for **Shopify** merchants, including **Lululemon**) and **AI document / invoice automation** (Sinoptics AI).

Core strengths: **Python** (including **FastAPI** where used), **AWS + GCP**, **API Gateway + Lambda** patterns deployed with **SST/CDK**, **Kubernetes** (GKE, Knative-style services, Helm), **relational + NoSQL** persistence, **Prometheus/Grafana** on telemetry Kubernetes (Helm), and **SLO-minded** operations (structured logging, Slack-driven ops alerts in infra templates, incident participation).

Constructor.io - Role Fit (Retail Media / Sponsored Listings)

Requirement	How your experience maps
Python + FastAPI for web services	FastAPI and Uvicorn on GCP (internal Knative-oriented API gateway service). FastAPI + Celery + Redis on K3S at Sinoptics. Python Lambda layers across multiple SST -deployed services (main public API gateway, billing, orders).
High-load / real-time	Telemetry backend: Knative-style HTTP services, Go/Gin and TypeScript , documented HTTP load-generator tests at high concurrency against telemetry endpoints. Spark / Airflow data plane at scale (see Data Engineer bullets below).
IaC - CloudFormation / Terraform	AWS SAM (Serverless Application Model) CloudFormation stacks for org-wide env + GitHub Actions deploy workflows. VPC stacks: CloudFormation + SAM configs. GKE hub: Helm plus documented Terraform workflow for related infra. GCP foundation: Pulumi (Python). Global webhook ingress: Pulumi entrypoint + GitHub Actions . SST + AWS CDK for Lambda/API stacks.
CI/CD (incl. GitHub Actions)	GitHub Actions on hub, webhook, and env stacks; Sinoptics and a separate public AI validation repo; AWS CodeBuild + CodeCommit and Google Cloud Build from broader platform work.
Observability (Prometheus / Grafana)	Telemetry codebase: Helm -based Prometheus (community chart) and Grafana datasource toward in-cluster Prometheus. Slack alert routing parameters in shared SAM/CloudFormation env templates. GCP Cloud Monitoring / Logging , AWS X-Ray on the main serverless API gateway.
MySQL / PostgreSQL, ClickHouse / Athena class	PostgreSQL / RDS (indexing and ops notes in internal custom-portal gateway docs). MongoDB , Redis , Neo4j , Spanner , BigQuery in Lambda Python layers. Lake / warehouse patterns: BigQuery , Hudi/Delta , BigLake .

SOA / microservices, API design	Many small deployable units: API Gateway + Lambda (SST/CDK), Pulumi webhook stack, Knative / kn func workflow in telemetry documentation, Helm -packaged hub service. OpenAPI-related Python tooling on the main gateway stack.
AWS / Azure / GCP	Deep AWS + GCP ; Azure not a primary production cloud in recent roles.
Cross-functional + English	GitBook docs, consulting product teams on data tooling, stakeholder communication in international teams.
Ad tech (ideal)	Not classical ad network - Shopify merchant surface, internal billing quota validation and billing flags Lambdas, ML gateway (SNS+SQS-triggered Python Lambda routing models, shadow writes to MongoDB).

Core Skills

Area	Tools and experience
Backend & APIs	Python (Lambda handlers + shared layers), FastAPI (GCP portal/Knative-oriented gateway, Sinoptics), Node.js / TypeScript in SST and telemetry codebases, REST behind API Gateway , Go/Gin (telemetry).
Cloud	AWS (Lambda, API Gateway, S3, SQS/SNS, EventBridge, RDS, Secrets Manager, EC2, ECR, CodeBuild, CodeCommit), GCP (GKE, BigQuery, GCS, Cloud Build, Artifact Registry, IAM / Workload Identity, Memorystore Redis).
IaC & config	Pulumi (Python) , AWS SAM / CloudFormation , SST + AWS CDK , Helm , Terraform (documented alongside GKE hub).
Datastores	PostgreSQL (RDS / portal gateway), MongoDB , Redis (incl. cluster client in layers), Neo4j , Spanner , BigQuery , lakehouse (Hudi / Delta , BigLake). MySQL in on-prem / Dify-style deployments (Sinoptics).
CI/CD & quality gates	GitHub Actions (hub, env stacks, webhook gateway; public side project: tests, Docker, Trivy SARIF), CodeBuild → ECR → Kubernetes , Cloud Build for data-plane images.
Orchestration & distributed work	Kubernetes (GKE, K3S/Rancher) , Airflow (pod-based tasks on GKE), Hatchet on K8s, n8n (Sinoptics), Knative tooling (kn / kn func in telemetry docs).
Observability & operations	Prometheus + Grafana (Helm) for telemetry Kubernetes; Slack channels wired through CloudFormation parameters; structured logging and pipeline alerting; DR patterns (secrets, encryption, backups).
E-commerce domain	Shopify integrations (official Shopify Python SDK and HMAC validation on main serverless gateway stack), Lululemon and enterprise merchant context, webhooks and order/product pipelines.

Professional Experience

Backend Engineer / AI Platform Engineer - Sinoptics AI

October 2025 - May 2026 (remote)

- Migrated **FastAPI + Celery + Redis** to **K3S (Rancher)** on bare-metal EC2 (AWS China); **hybrid CI/CD**: **GitHub Actions** → **CodeCommit** → **CodeBuild** → **ECR** → **kubectl rollout**, secrets in **Secrets Manager**.
- Ran **on-prem AI stack** (Dify, PostgreSQL, nginx, Let's Encrypt), **Hatchet** workflow engine on **Kubernetes** for **OCR / LLM** workloads.
- **Layered microservice architecture** (n8n → agents → aggregator) with **RBAC**, encryption, audit logs, **GDPR / CN** compliance awareness.

Backend Engineer - Yofi Inc. (USA)

February 2022 - October 2025 (remote)

Yofi: **anti-fraud / customer intelligence** for **Shopify** merchants (**Lululemon** and others). The bullets below reflect hands-on work across internal Yofi engineering codebases (API gateways, billing, orders, telemetry, shared env IaC, GKE hub, GCP foundation).

- **Serverless API surface (AWS)**: primary **SST + AWS CDK** stack for Customer, Order, Installation HTTP APIs - **API Gateway v2** constructs, **Lambda** (Node.js handlers plus **Python Lambda layer** with boto3, PyMongo, redis-py-cluster, Neo4j, BigQuery / Spanner / Firestore clients, **AWS X-Ray** SDK).
- **GCP portal / Knative path**: internal **FastAPI + Uvicorn** service with Kubernetes Python client, **Google Cloud Spanner**, **Redis** (Memorystore TLS), **GCP Secret Manager**, JWT admin APIs.
- **Custom portal gateway**: **SST v2 + AWS CDK Lambda Python** constructs, Jest tests; **PostgreSQL / RDS** indexing and tuning documented internally.
- **Global webhooks**: **Pulumi** (Python) stack, just + uv developer workflow, **GitHub Actions** for tests and deploy to dev/main → AWS.
- **Telemetry platform**: Knative-style services, **Go / TypeScript**, container images in ECR; **Prometheus** (community Helm chart) and **Grafana** datasource documentation; HTTP load-generator tests against public telemetry URLs.
- **ML routing**: SST + CDK; **SNS + SQS**-triggered **Python Lambda** selects model, writes **MongoDB** shadow predictions.
- **Billing controls**: SST + Python Lambda stacks for quota validation and billing flags; layers use boto3, PyMongo, **Redis** cluster client, Spanner / Google libs.
- **Orders**: SST + Python Lambda for order persistence and state prediction; PyMongo, Gremlin / graph-related dependencies in order persistence layers.
- **Shared hub on GKE**: **Helm** chart, Kubernetes deployment, OAuth2 with **Okta** for Airflow / Airbyte, **GitHub Actions** deploy workflows; documented **Terraform** workflow alongside hub infra.
- **Org-wide AWS env (IaC)**: large **SAM/CloudFormation** templates (Slack channel parameters for alerts); VPC stacks via CloudFormation + SAM.
- **GCP foundation**: **Pulumi** Python project; Cloud Run-style Python sources including Shopify webhook handlers wired through Pub/Sub.
- **Data plane at scale**: **Spark on Kubernetes**, **Airflow** (~25 prod DAGs), **GCS + BigLake + Hudi/Delta**, pipelines over **1B+** events.
- **Cross-team**: GitBook API docs, Airflow pod-task templates, training on DBT, Airbyte, Spark practices.

Representative stack: API Gateway, Lambda, SST, AWS CDK, SAM/CloudFormation, Pulumi, Terraform (documented), SQS/SNS, EventBridge, MongoDB, PostgreSQL, Redis, Neo4j, Spanner, BigQuery, Airflow, Spark, GKE, Helm, Docker, Prometheus, Grafana.

Head of IT and Finance - Engineering Solutions LLC

March 2013 - December 2017

- Led IT and finance; ERP, reporting, governance - stakeholder management and delivery under business constraints.

Head of IT Department - New Engineering Solution

April 2003 - February 2013

- Owned corporate systems and reporting; early data discipline and automation.

Open Source / Personal Templates

- **FastAPI on K3S** - FastAPI + Celery + Postgres + Redis, GitHub Actions → CodeBuild → ECR rollout (public template repo).
- **Diffy on Ubuntu, PostgreSQL on cloud VM** - repeatable installs, TLS, backups (public automation repos).
- **Airflow on GKE** - pod-based Spark/Python jobs, Cloud Build, dev/prod separation (public operator/template repo).

Side Project - AI Company-Validation (public repo)

Multi-agent company-validation and **invoice** workflows (n8n, Docker Compose, **MongoDB** logging). **GitHub Actions**: MongoDB + Redis service containers, Node + Python tests, Docker Compose build/smoke, **Trivy** SARIF upload, branch-based workflows.

Education

- **Southern Federal University** - PhD, Mathematical Sciences (*September 1994 - May 2003*)
- **RPA NEPA** - MBA, Information Technology (*September 2005 - May 2007*)

Certifications

- **Certified Accountant Practitioner (CAP)**
- Advanced training: **IFRS financial reporting, management accounting**

Additional

- Comfortable **remote**, async communication, **English** for design docs and incidents.
- Security- and **compliance-aware** delivery (fraud, finance, cross-border data).